Polarimetric SAR Time Series Change Analysis Over Agricultural Areas

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***Abstract*— This article proposes a change detection and analysis technique for monitoring the phenological development of agricultural vegetation by means of multitemporal Polarimetric Synthetic Aperture Radar (PolSAR) acquisitions. The technique relies on the generalized eigendecomposition of the polarimetric covariance matrices of the individual acquisitions. It both quantifies the magnitude of the change between PolSAR images acquired at different times and also provides an interpretation of occurred change in terms of the modified polarization states. This makes the algorithm suitable for investigating scattering dynamics associated with the phenological development of agricultural vegetation. To aid the interpretation of the changes detected, a representation based on the polarization states affected by the change process is proposed. The technique is evaluated using part of the multitemporal AGRISAR 2006 campaign data set. This data set consists of 12 quad-polarimetric images acquired by the German Aerospace Center (DLR) E-SAR airborne system at L-band from April 2006 to August 2006 over the Demmin test site. It covers large parts of the development cycle of different crop types. As a part of the evaluation, reference ground measurements are used to facilitate the interpretation of the data. The evaluation focuses on five important crop types: wheat, barley, rape, maize, and sugar beet. The results show that the proposed technique is able to detect and characterize different types of changes related to distinct development states of different crop types as the plant growing, maturation, and drying processes.**

***Index Terms*— Agriculture, change analysis, change detection, crop monitoring, polarimetry, synthetic aperture radar (SAR), time series.**

I. Introduction

SYNTHETIC aperture radar (SAR) configurations are active imaging systems that operate in the microwave range of the electromagnetic spectrum. They are able to provide high-resolution images of the radar reflectivity across large scenes [1]. Moreover, since the atmosphere is almost transparent at these frequencies, SAR systems are able to acquire imagery independently of the weather conditions or the day–night cycle. Furthermore, when operated on a spaceborne platform, they can achieve global coverage with a regular temporal resolution. In this sense, the increasing presence of spaceborne SAR sensors has facilitated the construction of temporal series data sets. These data sets contain different acquisitions of the same scene at different times. This information is of considerable value, as it allows us to monitor the temporal evolution or development of the scene. This trend is expected to continue in the future, as current and future missions focus on the systematic coverage of the Earth's surface at short repeat pass intervals. Relevant examples include the ESA Sentinel-1 or the future JAXA ALOS-4 and DLR Tandem-L missions. All this makes SAR systems a valuable source of information for the monitoring of Earth's environment.

SAR configurations are capable of achieving high spatial resolution by coherent processing of the radar echoes received by a moving platform. Even for high-resolution systems, however, the size of the resolution cell is much larger than the wavelength. The received echo for each resolution cell is given by the coherent combination of the responses of all the individual scatterers and depends on their distribution within the cell, which is generally unknown. As a consequence, in the absence of a dominant scatterer, the resulting image has a noisy appearance called *speckle* [2]. Speckle phenomenon complicates the interpretation of SAR data, especially in the case of change detection, as it hinders a reliable estimation of the target response. Numerous speckle reduction techniques have been proposed, which tends to improve the estimates by adapting to the content of the image. The employed methodologies range from a simple set of directional windows [3] to a more complex adaptive neighborhood [4] or a region-based approach [5]. In addition, some other nonlocal approaches have also been proposed, as described in [6], for instance.

Polarimetric SAR (PolSAR) systems provide additional information by exploiting the vectorial nature of the electromagnetic waves. When two orthogonal polarization states are employed in transmission and reception, polarization synthesis may be applied in order to explore the target response in the complete polarimetric domain [7], [8]. During the last decade, PolSAR has been effectively employed for extracting useful biophysical and geophysical information of the scene,

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which makes it very valuable and convenient for monitoring the Earth's surface [9]–[11].

Several methods have been proposed to aid in the interpretation of the obtained polarimetric response for the different areas of the scene. In this context, polarimetric decompositions are often used to decompose the target response information into a reduced set of parameters. Several polarimetric decompositions have been proposed, for instance: Huynen [12], Krogager [13], Cameron [14], Freeman and Durden [15], or Touzi [16] decompositions. One of the most popular decompositions, proposed by Cloude and Pottier [17], is based on the projection of the coherence matrix into the orthogonal polarimetric basis generated by its eigenvectors. A set of three parameters, entropy (H), anisotropy (A), and mean alpha angle ($\bar{\alpha}$), are obtained, which may be employed to interpret the scattering process. In this case, entropy indicates the power distribution of the different scattering processes, anisotropy indicates the dominance of the second eigenvalue versus the third one, in terms of the reflected power, and the mean alpha angle refers to the type of scattering [17].

Well-established approaches also exist for PolSAR change detection, as described in [18], [19], where a test statistic is employed to estimate the probability of change of the different areas of the image. Other techniques have proposed a similar concept in a region-based approach, as described in [20]. Although all these techniques detect the amount of change between two images, none of them provide information concerning the type of change. The change interpretation may be performed by analyzing a polarimetric decomposition of both images in order to see which parameters differ. However, this information is not complete, as only a reduced set of parameters is obtained from the complete polarimetric space. In addition, these decompositions describe the scattering for each of the images, not the change between them, which could cover-up some of the changes. This article proposes a fully polarimetric change analysis and decomposition technique to detect and analyze the changes between two images. A change representation for the identification and interpretation of the changes is proposed. This representation is generalized to time series, where it can be used to determine the principal changes in the development of different crop types.

The content of this article is structured as follows. Section II introduces the basics of SAR Polarimetry, focusing on the target description and its statistics. Section III analyzes different ways of detecting the changes between two PolSAR images and describes the proposed polarimetric change analysis and the corresponding change representation. In Section IV, the suggested technique is evaluated with a real data set over agricultural areas, focusing on the detection of the different crop changes over time. The results are compared with *in situ* collected measurements and photographs to evaluate the detected changes. Finally, Section V presents the conclusions of this article.

II. SAR Polarimetry

A PolSAR system measures the scattering matrix $\mathbf{S}$ for each resolution cell [8], [21]. Under the backscattering alignment convention, $\mathbf{S}$ may be vectorized into the scattering vector $\mathbf{k}$ in the Pauli basis [22] as

$$\mathbf{k} = \frac{1}{\sqrt{2}}[\, S_{hh} + S_{vv}, \quad S_{hh} - S_{vv}, \quad S_{hv} + S_{vh}\,]^T \tag{1}$$

where S_{ij} represents the complex scattering coefficient for the received and transmitted polarization combination $i, j \in [h, v]$, h and v denote the horizontal and vertical polarizations, respectively, and ^{T} stands for vector transpose.

If the number of individual scattering elements within the resolution cell is sufficiently large, then the central limit theorem applies [23], [24]. In this case, $\mathbf{k}$ follows a zero-mean complex Gaussian distribution

$$p_{\mathbf{k}}(\mathbf{k}) = \frac{1}{\pi^3 |\mathbf{T}|} \exp(-\mathbf{k}^H \mathbf{T}^{-1} \mathbf{k}) \tag{2}$$

where ^{H} represents the complex Hermitian transpose and $\mathbf{T}$ is the polarimetric coherency matrix, i.e., the covariance matrix in the Pauli basis, which parameterizes the distribution

$$\mathbf{T} = E\{\mathbf{k}\mathbf{k}^H\} \tag{3}$$

where $E\{x\}$ denotes the statistical expectation of the stochastic process x and $|\cdot|$ is the matrix determinant operator.

The polarimetric response is usually described by the sample coherency matrix $\mathbf{Z}$ that corresponds to the maximum likelihood estimator (MLE) of $\mathbf{T}$. Under the Gaussian hypothesis, it completely characterizes the polarimetric response [25]

$$\mathbf{Z} = \langle \mathbf{k}\mathbf{k}^H \rangle_n = \frac{1}{n} \sum_{i=1}^{n} \mathbf{k}_i \mathbf{k}_i^H \tag{4}$$

where $\mathbf{k}_i$ represents the scattering vector of the ith pixel and n represents the number of independent pixels averaged.

Note that this estimation is only valid when averaging homogeneous samples, i.e., samples with the same statistical distribution. In this case, the estimated sample coherency matrix $\mathbf{Z}$ is Wishart distributed, characterized by the distribution [25]

$$p_{\mathbf{Z}}(\mathbf{Z}) = \frac{n^{pn} |\mathbf{Z}|^{n-p}}{|\mathbf{T}|^n \widetilde{\Gamma}_p(n)} \text{etr}(-n\mathbf{T}^{-1}\mathbf{Z}) \tag{5}$$

where $\text{etr}(\mathbf{X})$ is the exponential of the trace of matrix $\mathbf{X}$ and

$$\widetilde{\Gamma}_p(n) = \pi^{\frac{1}{2}p(p-1)} \prod_{i=1}^{p} \Gamma(n - i + 1) \tag{6}$$

where $p = 3$ is the dimension of the $\mathbf{k}$ vector and Γ is the gamma function.

Note that the Wishart distribution defined in (5) is only valid for $n \geq 3$. For $n < 3$, the sample coherency matrix $\mathbf{Z}$ is singular, i.e., with a rank below 3 and $|\mathbf{Z}| = 0$. Therefore, some degree of averaging or speckle filtering is required for an appropriate polarimetric analysis.

III. Polarimetric Change Detection and Analysis

In the following, the sample coherency matrix $\mathbf{Z}$ will be employed assuming the validity of the Gaussian hypothesis for distributed scattering processes. The detection and analysis of changes between two polarimetric images of the same scene acquired at different times are based on the corresponding sample coherency matrices $\mathbf{Z}_1$ and $\mathbf{Z}_2$.

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A. Polarimetric Change Detection

The sample coherency matrix follows a Wishart distribution, as described in (5). This can be used to assess whether the two observed coherency matrices $\mathbf{Z}_1$ and $\mathbf{Z}_2$ correspond to the same distribution. This approach was employed in [18] in order to derive a Wishart test statistic Q

$$Q = 2^{6n} \frac{|\mathbf{Z}_1|^n |\mathbf{Z}_2|^n}{|\mathbf{Z}_1 + \mathbf{Z}_2|^{2n}} \tag{7}$$

where both sample coherency matrices $\mathbf{Z}_1$ and $\mathbf{Z}_2$ are assumed to be obtained by averaging the same number of samples n. Usually, the logarithmic test statistic $-\ln Q$ is employed, which has a large value when $\mathbf{Z}_1$ and $\mathbf{Z}_2$ belong to two different distributions and a small value otherwise. As long as $\mathbf{Z}_1$ and $\mathbf{Z}_2$ belong to the same distribution, it is assumed that no changes have occurred. On the other hand, a change in the scene will induce a change in the coherency matrix distribution.

Alternatively, the geometry of the Hermitian positive definite matrix cone may be used to define a similarity measure, that is, a geodesic measure adapted to the coherency matrix space [26]. This idea was employed in [20] to define a change measure across N acquisitions. Applying this measure to a pair of acquisitions, given by $\mathbf{Z}_1$ and $\mathbf{Z}_2$, the same d_g measure as defined in [26] is obtained

$$d_g = \left\| \log\left(\mathbf{Z}_1^{-1/2} \mathbf{Z}_2 \mathbf{Z}_1^{-1/2}\right) \right\|_F \tag{8}$$

where $\log(\mathbf{X})$ refers to the matrix logarithm and $\|\mathbf{X}\|_F$ represents the Frobenius matrix norm of $\mathbf{X}$. As for the abovementioned test statistic, this will yield lower values when $\mathbf{Z}_1$ and $\mathbf{Z}_2$ are similar to each other, indicating no change, and a large value otherwise.

B. Polarimetric Change Analysis

The measures defined in (7) and (8) may be employed to obtain an idea of the amount of change between $\mathbf{Z}_1$ and $\mathbf{Z}_2$, but they do not give information related to the type of change in the polarimetric space or the direction of change (i.e., increase or decrease).

One of the first approaches to analyze the change between two coherency matrices $\mathbf{Z}_1$ and $\mathbf{Z}_2$ was to employ the polarimetric contrast measure P_c at a given polarization state $\mathbf{w}$ [27], [28]

$$P_c(\mathbf{Z}_1, \mathbf{Z}_2, \mathbf{w}) = \frac{\mathbf{w}^H \mathbf{Z}_2 \mathbf{w}}{\mathbf{w}^H \mathbf{Z}_1 \mathbf{w}}. \tag{9}$$

This measure may be interpreted as the ratio between the backscattered power at both acquisitions for the polarization state $\mathbf{w}$. Since it is a ratio of two powers, P_c is always real and positive. $1 < P_c < \infty$ indicates an increase in backscattered power from $\mathbf{Z}_1$ to $\mathbf{Z}_2$ at the polarization state $\mathbf{w}$, whereas $0 < P_c < 1$ represents a decrease. When $P_c = 1$, there is no change.

The maximum (minimum) polarimetric contrast can be addressed in terms of the optimum polarization state $\mathbf{w}_{opt}$, which maximizes (or minimizes) P_c in (9). The solution for this optimization problem is related to the generalized eigenvalues of $\mathbf{Z}_1$ and $\mathbf{Z}_2$ [8], [27]

$$\mathbf{Z}_2 \mathbf{w} = \lambda \mathbf{Z}_1 \mathbf{w} \tag{10}$$

which may be obtained as the solution of the equation

$$|\mathbf{Z}_2 - \lambda \mathbf{Z}_1| = 0. \tag{11}$$

The generalized eigenvalues λ_i (for $i = 1, \ldots, 3$) define the P_c values that are obtained according to (9) using the corresponding generalized eigenvectors $\mathbf{w}_i$. Sorted in the descending order such that $\lambda_1 \geq \lambda_2 \geq \lambda_3 > 0$, λ_1 corresponds to the maximum and λ_3 to the minimum P_c, while λ_2 corresponds to an intermediate value.

If $\mathbf{Z}_1$ is invertible, then from (10) follows:

$$\mathbf{Z}_1^{-1} \mathbf{Z}_2 \mathbf{w} = \lambda \mathbf{w} \tag{12}$$

the classical eigenvalue problem for $\mathbf{Z}_1^{-1}\mathbf{Z}_2$ with the characteristic equation

$$\left| \mathbf{Z}_1^{-1} \mathbf{Z}_2 - \lambda \mathbf{I} \right| = 0. \tag{13}$$

It is worth noticing that λ_i and $\mathbf{w}_i$ depend on both covariance matrices $\mathbf{Z}_1$ and $\mathbf{Z}_2$. They represent only the relative changes between them according to the ratio defined in (9). Since the $\mathbf{Z}_1^{-1}\mathbf{Z}_2$ matrix is in general not Hermitian, the generalized eigenvectors $\mathbf{w}_i$ are not orthogonal to each other, unlike the eigenvectors of $\mathbf{Z}_1$ and $\mathbf{Z}_2$. If both matrices are not singular, then the set of $\mathbf{w}_i$ vectors confirm a nonorthogonal basis for the complete polarimetric space [29], [30].

The relation between the generalized eigenvalues and the polarimetric contrast was already analyzed in [31] for polarimetric change detection, where the generalized eigenvalues λ_i were used to obtain the magnitude of the change. Here, the information of the generalized eigenvectors $\mathbf{w}_i$ is exploited for the interpretation of the type of change. This interpretation is based on the fact that any unitary vector can be associated with a scattering matrix, i.e., a rank 1 scatterer [8], [32]. While the obtained $\mathbf{w}_i$ is not restricted by orthogonality, it is important to note that this restriction $\mathbf{w}_i^H \mathbf{w}_j = 0$ for $i \neq j$ is not broken but replaced by [29]

$$\mathbf{w}_i^H \mathbf{Z}_2 \mathbf{w}_j = 0 \quad \forall i \neq j$$
$$\mathbf{w}_i^H \mathbf{Z}_1 \mathbf{w}_j = 0 \quad \forall i \neq j.$$

In fact, the generalized eigenvectors $\mathbf{w}_i$ conform a nonorthogonal basis that simultaneously diagonalizes both $\mathbf{Z}_1$ and $\mathbf{Z}_2$ Hermitian matrices [33]. Formally, they correspond to the projection vectors that maximize and minimize (9).

When a temporal series of N coherency matrices $\mathbf{Z}_1, \mathbf{Z}_2, \ldots, \mathbf{Z}_N$ acquired at different times is available, the change analysis can be performed between every acquisition pair $\mathbf{Z}_i, \mathbf{Z}_j$. In this case, the obtained change contributions are interconnected. For instance, when considering three acquisitions, the change from the first to the second and from the second to the third acquisition is obtained from the eigendecomposition of the matrices $\mathbf{M}_{1\to2} = \mathbf{Z}_1^{-1}\mathbf{Z}_2$ and $\mathbf{M}_{2\to3} = \mathbf{Z}_2^{-1}\mathbf{Z}_3$, as described in (12). This also implies that

$$\mathbf{Z}_2 = \mathbf{Z}_1 \mathbf{M}_{1\to2} \tag{14}$$

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so that the change between the first and the third acquisitions $\mathbf{M}_{1\to3}$ is related to $\mathbf{M}_{1\to2}$ and $\mathbf{M}_{2\to3}$ as

$$\begin{aligned}\mathbf{Z}_3 &= \mathbf{Z}_2\mathbf{M}_{2\to3} = \mathbf{Z}_1\mathbf{M}_{1\to2}\mathbf{M}_{2\to3}\\ \mathbf{M}_{1\to3} &= \mathbf{Z}_1^{-1}\mathbf{Z}_3 = \mathbf{M}_{1\to2}\mathbf{M}_{2\to3}.\end{aligned} \tag{15}$$

In general, for each i, j pair such that $j > i$, it can be seen that $\mathbf{Z}_j = \mathbf{Z}_i\mathbf{M}_{i\to j}$ where

$$\mathbf{M}_{i\to j} = \mathbf{Z}_i^{-1}\mathbf{Z}_j = \prod_{k=i}^{j-1}\mathbf{M}_{k\to k+1}. \tag{16}$$

For this reason, this change analysis technique is referred to as multiplicative change analysis [31], [34], [35], in contrast to the additive change analysis [36], where changes are defined in an additive context $\mathbf{Z}_2 = \mathbf{Z}_1 + \mathbf{C}_{1\to2}$.

C. Polarimetric Change Representation

Following the abovementioned discussion, the generalized eigenvalues and eigenvectors can be used to represent the change between $\mathbf{Z}_1$ and $\mathbf{Z}_2$. To this end, the generalized eigenvectors $\mathbf{w}_i$ normalized such that $\|\mathbf{w}_i\| = 1$ and expressed in the Pauli basis are weighted by the associated generalized eigenvalues λ_i in decibel [34]. In addition, the changes are separated into increasing ($\lambda_i > 1$) and decreasing ($\lambda_i < 1$) in terms of power ratio. In this way, two 3-D real vectors $\mathbf{p}_{\text{inc}}$ and $\mathbf{p}_{\text{dec}}$ comprising the magnitude of the Pauli elements of $\mathbf{w}_i$ weighted by their significance are obtained, $\mathbf{p}_{\text{inc}}$ associated with the increasing and $\mathbf{p}_{\text{dec}}$ with the decreasing scattering contributions. Accordingly, each element k of $\mathbf{p}_{\text{inc}}$ and $\mathbf{p}_{\text{dec}}$, and p_{inc}^k and p_{dec}^k is defined as [34], [35]

$$p_{\text{inc}}^k = \left[\sum_{i|\lambda_i>1}^{p}\left(10\log_{10}\lambda_i\cdot|w_i^k|\right)^2\right]^{\frac{1}{2}} \tag{17}$$

$$p_{\text{dec}}^k = \left[\sum_{i|\lambda_i<1}^{p}\left(-10\log_{10}\lambda_i\cdot|w_i^k|\right)^2\right]^{\frac{1}{2}} \tag{18}$$

which can also be expressed in vector notation as

$$\mathbf{p}_{\text{inc}} = \left[\sum_{i|\lambda_i>1}^{p}(10\log_{10}\lambda_i)^2\cdot\mathbf{w}_i\odot\mathbf{w}_i^*\right]^{\odot\frac{1}{2}} \tag{19}$$

$$\mathbf{p}_{\text{dec}} = \left[\sum_{i|\lambda_i<1}^{p}(-10\log_{10}\lambda_i)^2\cdot\mathbf{w}_i\odot\mathbf{w}_i^*\right]^{\odot\frac{1}{2}} \tag{20}$$

where $\odot$ represents the element by element product (the Hadamard product) and $\mathbf{a}^{\odot1/2}$ represents the element by element square root of vector $\mathbf{a}$.

Even if $\mathbf{p}_{\text{inc}}$ and $\mathbf{p}_{\text{dec}}$ contain only a subset of the information provided by the generalized eigenvalue decomposition in (10), since the complex phases between different polarimetric channels are ignored, they still provide a useful visualization of the magnitude and type of change occurred. In terms of an RGB representation, it combines the (increasing and decreasing) change information for all the λ_i and $\mathbf{w}_i$ into two images where the amount of change is represented by the color brightness, while the color itself indicates the type of change.

TABLE I

LIST OF E-SAR ACQUISITIONS OVER THE DEMMIN TEST SITE IN THE AGRISAR 2006 CAMPAIGN. ALL ACQUISITIONS WERE COLLECTED AT X-, C-, AND L-BANDS IN SINGLE-POL, DUAL-POL, AND QUAD-POL, RESPECTIVELY

Acq. Date	Heading	Notes
19/04/2006	270°	Intensive ground measurements
20/04/2006	203°	Intensive ground measurements
03/05/2006	270°	
11/05/2006	270°	
16/05/2006	270°	
24/05/2006	270°	
06/06/2006	203°	Intensive ground measurements
07/06/2006	270°	Intensive ground measurements
13/06/2006	270°	
21/06/2006	270°	
05/07/2006	270°	Intensive ground measurements
06/07/2006	203°	Intensive ground measurements
12/07/2006	270°	
26/07/2006	270°	
02/08/2006	270°	

IV. EXPERIMENTAL RESULTS

For the evaluation of the proposed technique, a subset of the data acquired in the frame of the AgriSAR 2006 campaign [37] is used. The campaign took place from April 2006 to August 2006 on the Demmin test site located in Northern Germany. Demmin is mainly an agricultural site with fields of different crop types. During this time, DLR's airborne E-SAR sensor acquired, almost every two weeks, fully polarimetric SAR data at L-band as well as dual-pol and single-pol data at C- and X-bands, respectively. Parallel to the SAR acquisitions, a large set of ground measurements, including the documentation of the phenological development stage, vegetation height, and plant water content, as well as soil moisture, has been collected at every date. On the three "intensive" campaign dates, i.e., April 18–20, 2006, June 5–8, 2006, and July 3–8, 2006, a higher sampling of ground measurements was performed, and the supplementary information was collected. On the intensive campaign dates, the SAR data acquisition was also performed along with two different headings (270° and 203°). The list of SAR acquisitions and the dates corresponding to the intensive ground measurements are summarized in Table I. In the following, the 12 L-band acquisitions performed at a single heading (270°) will be used for the change analysis evaluation.

A. Results of the Three Intensive Measurement Acquisitions

We start with the change analysis between the three intensive campaign dates. In order to reduce the effect of speckle and to obtain a reliable estimation of the polarimetric response, the temporal evolution binary partition tree (TE BPT) with a pruning threshold of $\delta_p = -2$ dB has been employed, as described in [20].

The Pauli RGB images of these three acquisitions are shown in Fig. 1, while Fig. 2 shows the corresponding land-use map where the different colors represent the different crop types. Note that the large variation of the incidence angle may be observed from near to far range typical for airborne

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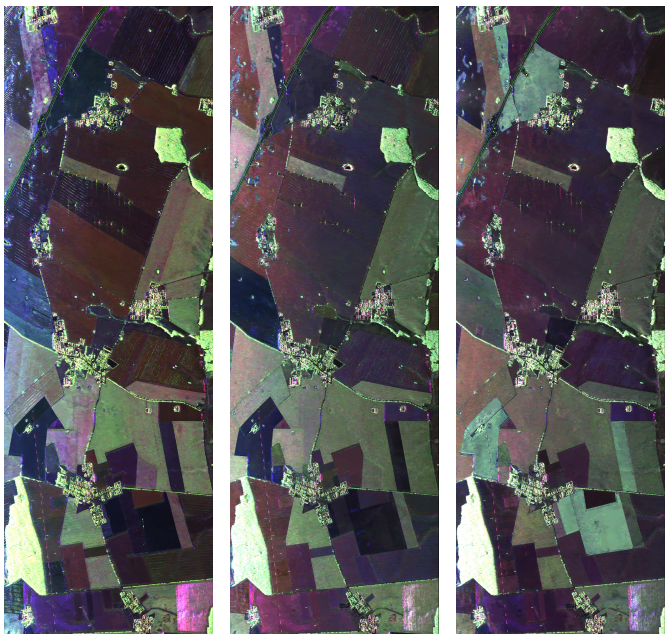

(a) (b) (c)

Fig. 1. Pauli RGB of the three L-band acquisitions of the AgriSAR 2006 campaign processed with the temporal evolution BPT ($\delta_p = -2$ dB). (a) 19th of April. (b) 7th of June. (c) 5th of July.

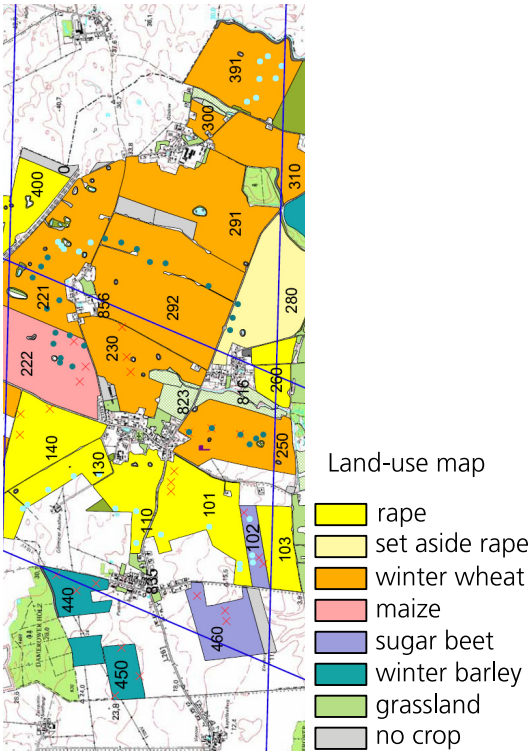

Fig. 2. Land-use map of the imaged area.

acquisitions (in our case from 25° to 55°), which entails the dependence of the scattering mechanisms on the angle of incidence; even for the same crop type, a significant change in the polarimetric signature from near to far range may occur.

The change of the vegetation height and plant water content during the campaign is shown in Fig. 3, where the three intensive campaign dates are marked by a red rectangle. For

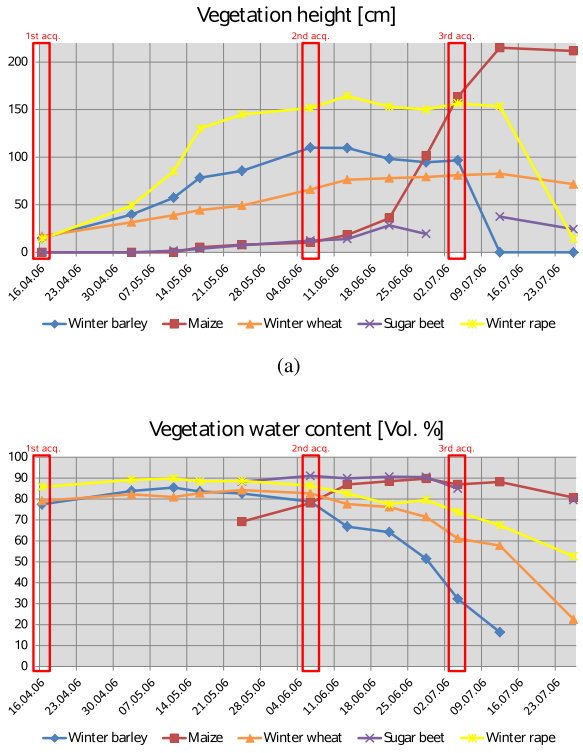

Fig. 3. Vegetation height and water content evolution for different types of crops. (a) Vegetation height. (b) Plant water content.

the evaluation of the phenological stage of the individual crops, the BBCH scale (Biologische Bundesanstalt Bundessortenamt und CHemische Industrie) [38] is used. The BBCH scale differentiates between ten main growth stages (indexed from 0 to 9). Within each main growth stage, further ten secondary development stages (indexed as well from 0 to 9) are differentiated so that each field is denoted by an index ranging from 00 to 99, characterizing its phenological stage as follows:

- 00–09: Germination.
- 10–19: Leaf development.
- 20–29: Formation of side shots/tillering.
- 30–39: Stem elongation or rosette growth.
- 40–49: Development of harvestable vegetative plant parts or vegetatively propagated organs/booting.
- 50–59: Inflorescence emergence/heading.
- 60–69: Flowering.
- 70–79: Development of fruit.
- 80–89: Ripening or maturity of fruit and seed.
- 90–99: Senescence.

Fig. 4 shows the BBCH scale for the crop types in the site during the campaign.

The change analysis between the three "intensive" campaign dates is shown in Fig. 5. Fig. 5(a) shows the results of the Wishart test statistic $-\ln Q$ measure, with Q defined in (7), for the acquisitions on the 19th of April and the 7th of June. In addition, Fig. 5(b) and (c) shows the change according to the d_g measure defined in (8) between the 19th of

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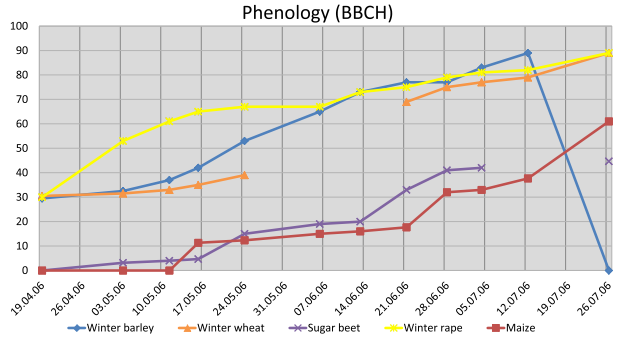

Fig. 4. Phenology evolution measured from ground according to the BBCH scale.

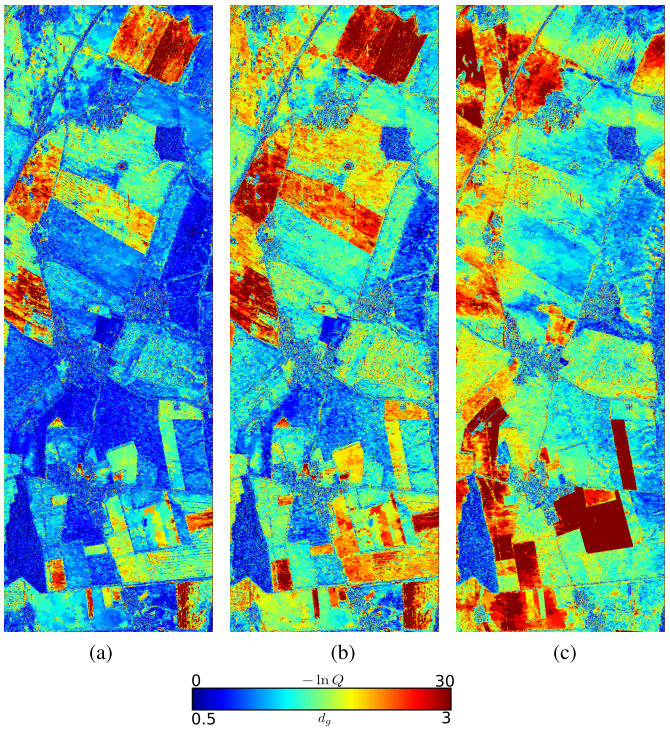

Fig. 5. (a) Wishart test statistic $-\ln Q$ (7) measure and (b) geodesic similarity measure d_g (8) change detection maps for the acquisition pair 19th of April—7th of June and d_g for acquisition pair 7th of June–5th of July (c). No change is represented in blue and a large change in red. (a) $-\ln Q$, 19/4–7/6. (b) d_g, 19/4–7/6. (c) d_g, 7/6–5/7.

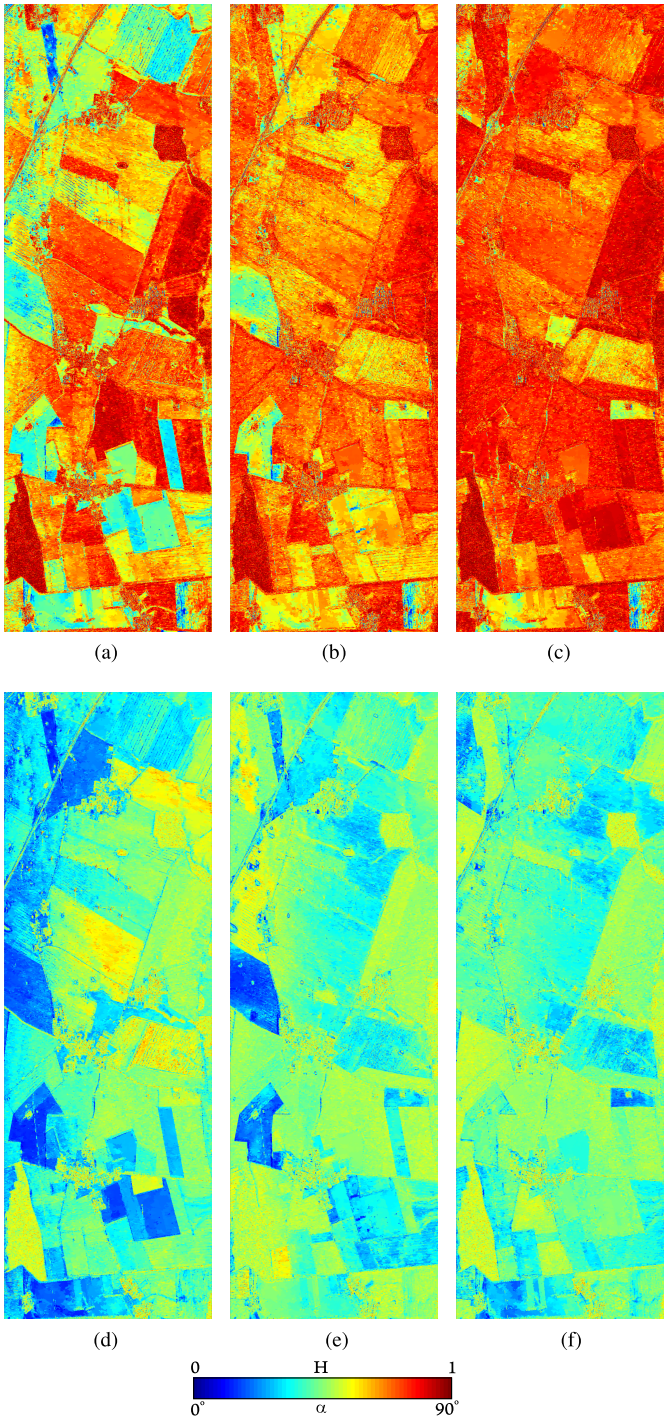

Fig. 6. Entropy (H) and averaged alpha angle ($\bar{\alpha}$) eigendecomposition parameters for the three selected acquisitions. (a) H 19th of April. (b) H 7th of June. (c) H 5th of July. (d) $\bar{\alpha}$ 19th of April. (e) $\bar{\alpha}$ 7th of June. (f) $\bar{\alpha}$ 5th of July.

April and the 7th of June and between the 7th of June and the 5th of July.

For both change metrics, a larger value indicates a bigger change, appearing in reddish colors. On the other hand, bluish regions represent areas that have not changed significantly.

As mentioned in Section I, polarimetric decomposition techniques may be employed to interpret the scattering occurring in each of the acquisitions [39]. For comparison, the entropy (H) and the mean alpha angle ($\bar{\alpha}$) [17] for the three acquisitions are shown in Fig. 6. Some of the changes detected in Fig. 5 may be also depicted when looking at the change of H and $\bar{\alpha}$, but many others not. This is especially the case between the second and the third acquisition, where the mean alpha angle remains the same over most of the fields and only small changes of the entropy are visible. As a result, the detection and interpretation of changes may not be easy using these parameters. This is due to the fact that they rather provide information related to the type of scattering at each acquisition than about its change between acquisitions.

In order to obtain the amount and type of change, the polarimetric change analysis proposed in Section III-B is applied. The generalized eigenvalues for the acquisitions on the 19th of April and the 7th of June are shown in Fig. 7. The eigenvalues are sorted ($\lambda_1 \geq \lambda_2 \geq \lambda_3$) and scaled in dB. No change

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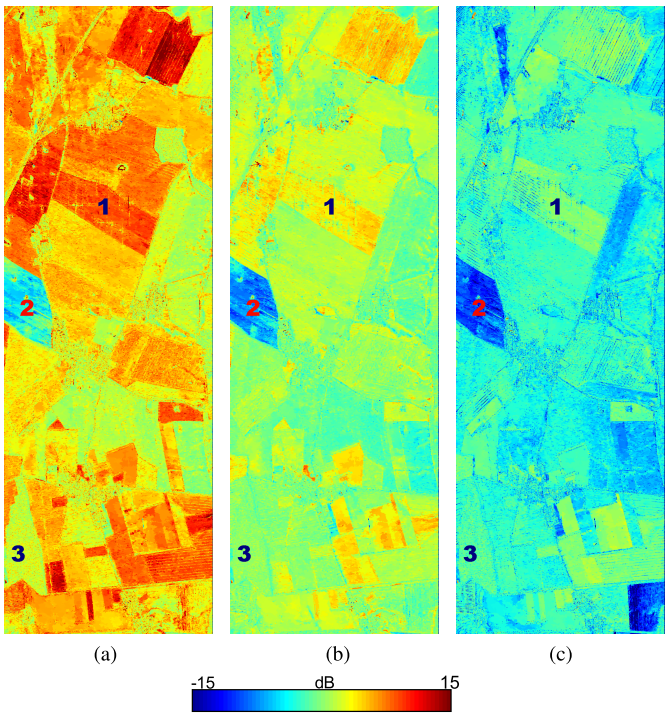

Fig. 7. Magnitude representation of the three generalized eigenvalues (in dB) depicting the change between the acquisitions from 19th of April to 7th of June. (a) λ_1. (b) λ_2. (c) λ_3.

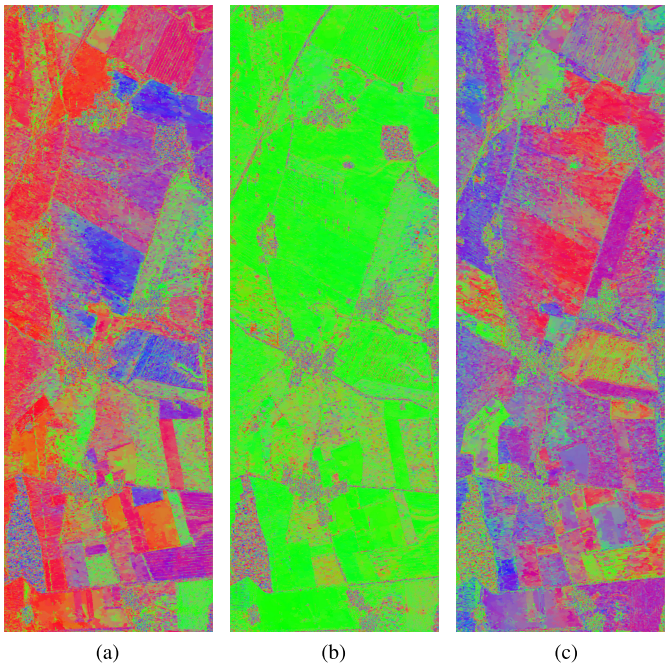

Fig. 8. Pauli representation of the three generalized eigenvectors depicting the change between the acquisitions from 19th of April to 7th of June. (a) $\mathbf{w}_1$. (b) $\mathbf{w}_2$. (c) $\mathbf{w}_3$.

corresponds to 0 dB, while the increase or decrease appears with positive or negative values, respectively. The Pauli RGB representations of the associated generalized eigenvectors are shown in Fig. 8.

Comparing the change detection results in Fig. 5(b), with the corresponding generalized eigenvalues, in Fig. 7, makes the correlation between the areas affected by large changes

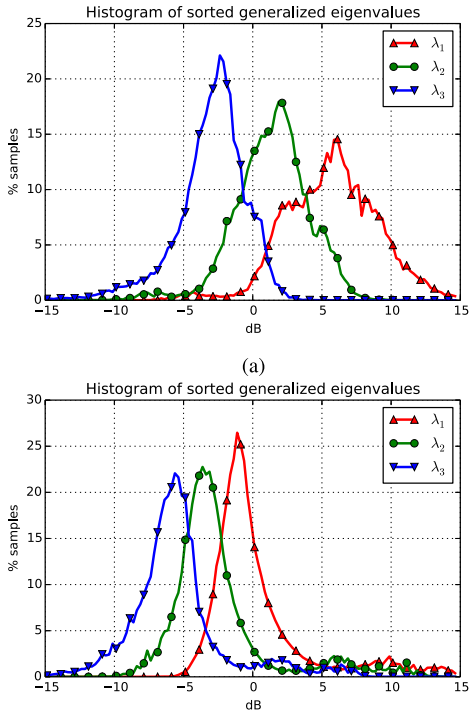

Fig. 9. Histograms of the generalized eigenvalues λ_i. (a) From 19th of April to 7th of June. (b) From 7th of June to 5th of July

obvious. In fact, since the change measures in Figs. 5(a) and 10 do not distinguish between increase or decrease, they may correspond to an increase ($\lambda > 0$ dB) or a decrease ($\lambda < 0$ dB) of the backscattered power and appear reddish in Fig. 7(a) or bluish in Fig. 7(c), respectively.

Fig. 9 shows the histograms of the generalized eigenvalues for both dates, expressing the overall behavior of the polarimetric change. As it may be seen in Fig. 9(a), there is a wide range of changes occurring between the 19th of April and 7th of June. The peak of λ_1 is at about 6 dB, while the peak of λ_3 is at -2.5 dB, indicating a relevant increase as well as a decrease in the power of individual scattering contributions. However, most of the λ_i values are larger than 0 dB, indicating that the increasing changes are dominating. The changes occurring in the time between the 7th of June and 5th of July have different behavior, as shown in Fig. 9(b). The peaks of all the histograms are negative, indicating a decrease in the power of the scattering contributions for most of the areas. Moreover, the shape of the histograms is also different, with narrower distributions in Fig. 9(b) than in Fig. 9(a).

It is important to note here that the full change information is expressed by the full set of generalized eigenvalues (as shown in Fig. 7) and eigenvectors (as shown in Fig. 8). For example, the wheat field marked with "1" in Fig. 7 is characterized by a strong increase in λ_1 [see Fig. 7(a)] associated with a change in $\mathbf{w}_1$ [as shown in Fig. 8(a)] but no significant change in λ_3 [see Fig. 7(c)]. On the contrary, the maize field marked with "2" in Fig. 7 shows the strongest change (a decrease) in λ_3 associated with the polarization state $\mathbf{w}_3$. Fields that do not change significantly are characterized by generalized eigenvalues close to 0 dB and appear in green in

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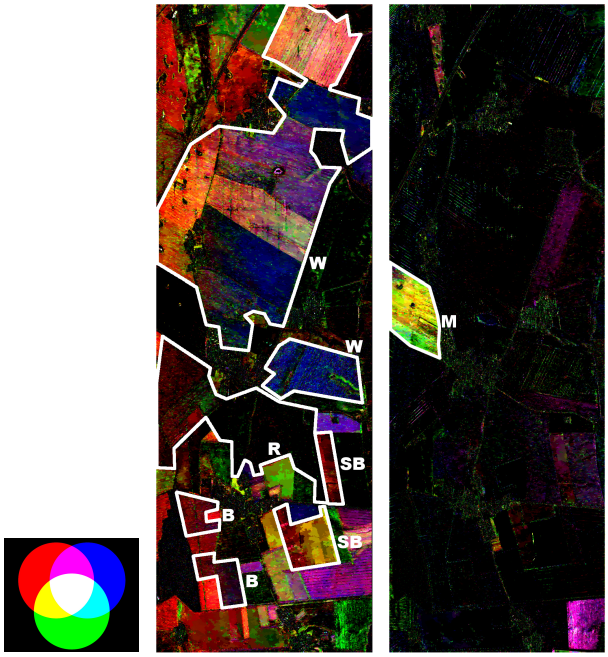
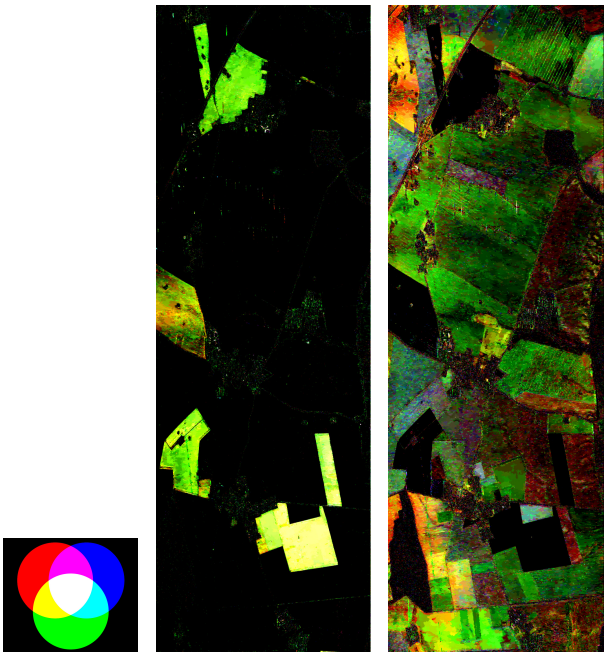


(a) (b)

Fig. 10. Polarimetric change representation for differences from 19th of April to 7th of June acquisitions. The labels represent the crop type: (W) wheat, (B) Barley, (R) rape, (M) maize, and (SB) sugar beet. (a) $\mathbf{p}_{\text{inc}}$. (b) $\mathbf{p}_{\text{dec}}$.



(a) (b)

Fig. 11. Polarimetric change representation for differences from 7th of June to 5th of July acquisitions. (a) $\mathbf{p}_{\text{inc}}$. (b) $\mathbf{p}_{\text{dec}}$.

all λ_i images. Such a case is the forested area marked with "3" in Fig. 7. For these no-change areas, the obtained generalized eigenvectors (see Fig. 8) are noisier corresponding to variations induced by speckle. This is in accordance with the results discussed in [40]. In conclusion, the change interpretation might only be achieved by combining the information provided by the generalized eigenvalues and eigenvectors, as shown in Figs. 7 and 8.

In order to consider all this information, the change representation proposed in Section III-C weights the contribution of each generalized eigenvector $\mathbf{w}_i$ according to the magnitude of the change given by the corresponding generalized eigenvector λ_i. This representation combines the (increasing and decreasing) change information for all the λ_i and $\mathbf{w}_i$ into two images. Figs. 10 and 11 show the change representation proposed in (17) and (18), i.e., the Pauli representation of the weighted polarization states in which the changes are observed. Both RGB images are scaled from 3 to 10 dB. This scaling filters out the small changes introduced by speckle as all areas with an intensity change below $\pm$ 3 dB appear in black.

Fig. 12 shows photographs of the relevant crop fields acquired at the three intensive campaign days. The numbers correspond to the field numbers used in the land-use map shown in Fig. 2. The changes relevant for the interpretation of Fig. 10 are between the first and second rows of Fig. 12. Similarly, the changes appearing in Fig. 11 correspond to the differences between the second and third rows of Fig. 12.

Looking at Fig. 10, a large number of fields appear in the $\mathbf{p}_{\text{inc}}$ image, indicating an increase, mostly in red and blue colors. According to the ground measurements shown in Fig. 3, this is mainly due to the plant growth, as depicted in the vegetation height plot in Fig. 3(a). Especially for wheat, barley, and rape, a clear increase in vegetation height is given for most of the fields where the increase is detected in Fig. 10(a). For sugar beet and maize crops, a smaller increase in vegetation height is observed as they are planted later. This is also clearly indicated by the BBCH phenology plot in Fig. 4. The changes on the sugar beet and winter barley fields, that appear in red and yellow colors in Fig. 10(a), are shown in Fig. 12(c)–(i) and (f)–(l), respectively.

The rape fields do not follow this trend, even if they are characterized by a significant increase in the height, as indicated in Fig. 3(a) and shown in Fig. 12(d)–(j). Only for a single field located in the near range, i.e., at the left part of the changing image in Fig. 10(a), an increase is visible. All other rape fields appear black, indicating that the polarimetric change is below $\pm$ 3 dB. The reason for this is probably in the dominant volume scattering contribution that characterizes the rape fields in both acquisitions: the plant density, the wet biomass, and the water content [see Fig. 3(b)] is larger for the rape than for other types of crops in the scene. This increases volume scattering and limits, at the same time, the penetration of the radar waves through the vegetation to the ground, preventing that the change in vegetation height is translated into a change in the polarimetric response.

For the winter wheat fields, a strong increase is detected in Fig. 10(a), associated with the plant growth, as shown in the plot in Figs. 3(a) and 12(a)–(h). However, it is remarkable that some of the fields appear reddish, while others bluish although no clear difference is observed between these fields. The different colors indicate that the change occurs at different polarizations, i.e., affects different scattering contributions, which may be induced by the fact that these fields are in a different phenological state. Unfortunately, the BBCH

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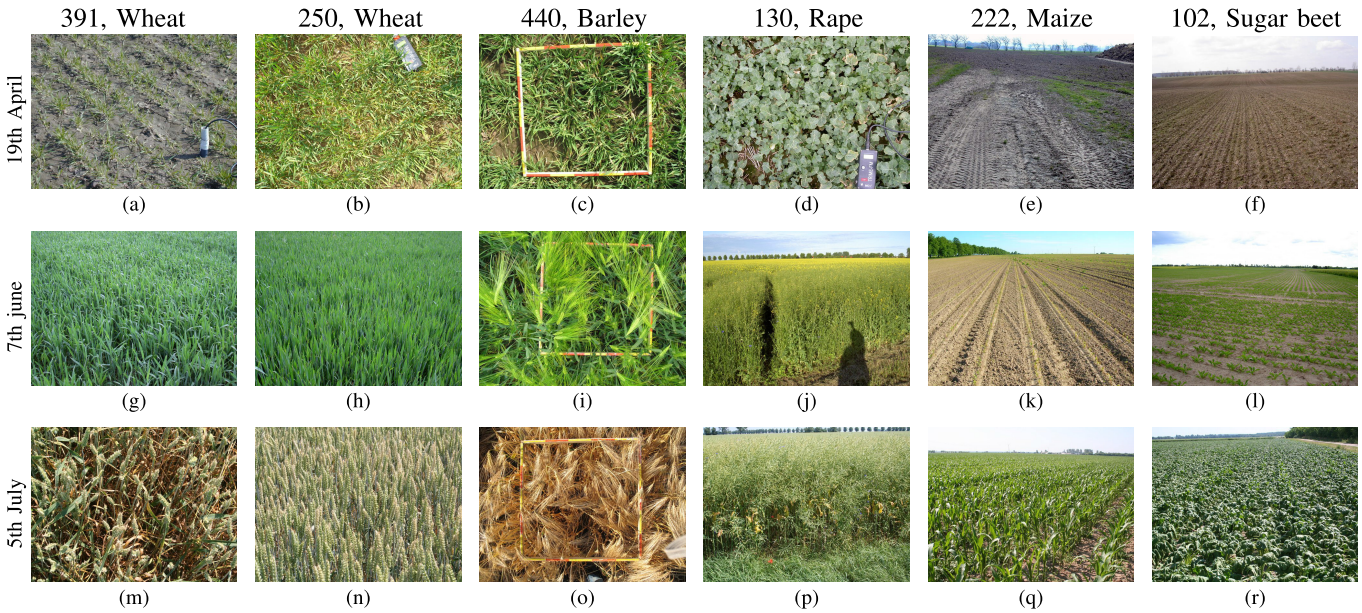

Fig. 12. Photographs of the different fields discussed. The field number corresponds to the land-use map in Fig. 2. The columns represent different fields, as indicated on top, while the rows indicate the three intensive measurement dates, as indicated in the left.

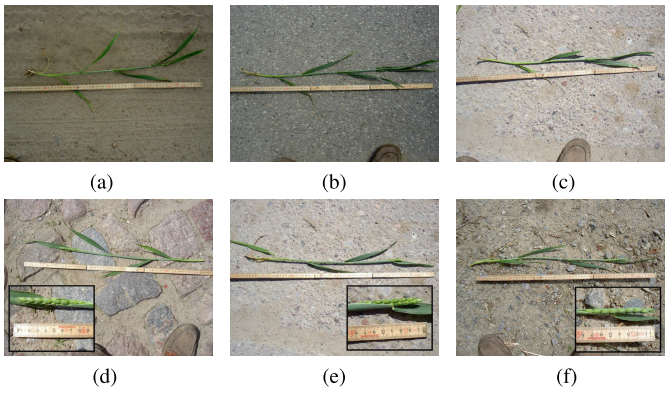

Fig. 13. Photographs of the different winter wheat plants on 7th of June. A zoomed-in view is shown in the black rectangle for those plants that have started the heading process in (d)–(f). (a) Field 391. (b) Field 291. (c) Field 221. (d) Field 310. (e) Field 292. (f) Field 250.

measurements in Fig. 4 represent only the average crop type evolution and are not specific for each field. Moreover, there are missing measurements for the wheat crop at these dates. Fortunately, during the intensive measurement campaigns, a large number of photographs were collected, which allows to visualize these differences. Some photographs of the wheat plants on the distinct fields are shown in Fig. 13. They make clear that the fields appearing in red in Fig. 10(a) have not started with the heading process [see Fig. 13(a)–(c)], whereas the fields appearing in blue color have started this process [see Fig. 13(d)–(f)].

Note that this does not mean that the method is able to detect the heading process of the plant directly. It rather means that the fields have a different polarimetric change between these two acquisitions, probably due to the different phenological evolution that alters their polarimetric response in both acquisitions.

Fig. 11 shows the polarimetric change maps associated with the changes occurring between the 7th of June and the 5th of July. Compared with the change maps associated with the changes occurring between the 19th of April and the 7th of June (see Fig. 10), they reflect a different change behavior, dominated by an intensity decrease for most of the fields. The ground measurement in Fig. 3 indicates that two types of change are ongoing between these dates. For maize and sugar beet, an increase in vegetation height is observed, while the plant water content remains constant. Note that these crops grow later, compared with the other crops, as it may be seen in Fig. 4. The discussed changes may also be observed in the photographs in Fig 12. The growth is also documented by Fig. 12(k)–(q) for the maize and Fig. 12(l)–(r) for the sugar beet. Accordingly, these fields appear as increasing intensity changes in green (and some red) colors in Fig. 11(a). For all other crops in the scene (i.e., winter wheat, rape, and winter barley), Fig. 3 indicates that the vegetation height remains constant, but the water content decreases. These crops are at the time of the second acquisition (7th of June) already grown up, and then, in the time between the second and the third (5th of July) acquisitions, they undergo the plant maturation and the drying out at the end of the phenological stage. Especially, the plant drying is associated with a decrease in the backscattered power clearly visible in the photographs in Fig 12, indicated by the yellowish color for the winter wheat [see Fig. 12(g)–(m)], rape [see Fig. 12(j)–(p)], and especially winter barley [see Fig. 12(i)–(0)]. However, for the different crops, the changes imposed by the plant drying appear in $\mathbf{p}_{\text{dec}}$ [see Fig. 11(b)] with different colors, i.e., are associated with different scattering mechanisms. The winter wheat fields appear in green (indicating a decrease in volume scattering), whereas a more yellowish color is obtained for the winter barley crops (i.e., decrease in the volume and dihedral). In the case of the rape, the color depends on the incidence angle; the rape fields in the left part of the image (in near range) appear more greenish (decrease in the volume scattering). This may be because of the larger penetration associated with the steeper

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incidence angles. Note that these are the same fields where the plant growth was detected in Fig. 10(a). The rape fields on the right part of the map (i.e., in far range) appear reddish (indicating a decrease in dihedral scattering).

Finally, it is important to mention that most of the discussed changes appearing in the change representation images shown in Figs. 10 and 11 may also be detected in the Pauli RGB images and the entropy and mean alpha angle maps, as shown in Figs. 1 and 6, respectively. However, their identification and interpretation are either more vague and unclear or not always possible.

The polarimetric change representation for the acquisitions from the 19th of April to the 5th of July will not be discussed in this section. However, it can be inferred from the previous two analyses. From a mathematical point of view, the polarimetric change analysis will result in the multiplicative combination of the previous two, as previously indicated in (15). Alternatively, Section IV-B describes a systematic method for change analysis in time series.

B. Results Over All the Acquisitions

The AgriSAR 2006 campaign comprises a time series of 12 L-band fully polarimetric acquisitions at the same geometry, from the 12th of May to the 4th of August. This allows 66 combinations of a different acquisition pairs for change analysis. The previous approach, then, becomes infeasible for the individual analysis of the $\mathbf{p}_{\text{inc}}$ and $\mathbf{p}_{\text{dec}}$ for each of those pairs. Note that although the different acquisition pairs are related according to (16), the full picture of all the change pairs may give a valuable idea of how successive changes combine with each other. Therefore, for the analysis of the changes among all the acquisitions of the data set, a different approach is required. A change matrix representation is proposed, which contains the results of the previously mentioned change representation between all the acquisition pairs [41]. Instead of comparing the results pixel by pixel, a set of regions is defined over the image, corresponding to homogeneous areas of the agricultural fields, and the change analysis is performed over the average covariance matrices in these regions at different dates. The horizontal and vertical axes of the change matrix correspond to the acquisition dates of each image. For each combination of acquisition pairs (i, j), in the upper diagonal, the $\mathbf{p}_{\text{inc}}$ representation between the i and j acquisitions is shown, whereas in the lower diagonal (j, i), $\mathbf{p}_{\text{dec}}$ between i and j is displayed.

Fig. 14 presents a scheme of the change matrix and depicts the main phenological and/or development change signatures for agricultural crops that may be detected in different colors, according to the results already analyzed in Section IV-A.

1) Seedbed preparation and planting [see Figs. 12(e)–(k)]. In this case, a decrease in the surface roughness due to the seedbed preparation is translated into a decrease in the backscattered signal, appearing in the lower diagonal of the change matrix.
2) The plant growing process [see Fig. 12(a)–(g)], corresponding to phenological stages 30–59 in the BBCH scale, including stem elongation, booting, and heading. When the plant grows, an increase in the radar backscattering signal is obtained due to the increased reflectivity of the plant. In this case, then, it appears in the upper diagonal of the change matrix.
3) The fruit development [see Fig. 12(g)–(m)], corresponding to phenological stages 60–79 in the BBCH scale, related to the flowering and fruit development of the plant. After the growing process, the plant develops its fruit, and a moderate reduction in water content is produced, which is perceived as a decrease in the $\mathbf{p}_{\text{dec}}$ change representation.
4) The plant drying [see Fig. 12(i)–(o)], corresponding to the BBCH scale stages 80–99, including ripening and senescence. In the final stages of the plant before harvesting, a strong decrease in water content is observed, which is translated into a strong decrease in the backscattering.
5) The crop harvesting, associated with a strong decrease in the volume scattering due to the removal of most of the plant. However, an additional increase may also be seen in different polarization states, representing the increase in the ground scattering component. In some cases, the lower part of the plant is left over the field after harvesting, contributing also to this increase in the ground scattering. Accordingly, harvesting may be mapped in the lower but also in the upper diagonal block of the change matrix.

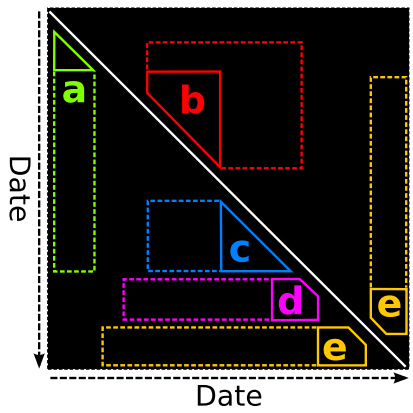

Fig. 14. Scheme of the change matrix and the main expected detected changes. (a) Seedbed preparation and planting. (b) Plant growing. (c) Fruit development. (d) Plant drying. (e) Harvest.

For the interpretation of the change matrix representation, it is important to keep in mind that the amount and nature of the change processes represented at each entry also depend on the temporal distance between the associated acquisitions. In general, for temporally adjacent acquisitions, only one type of change occurs, assuming a sufficiently high temporal sampling of the ongoing change process. These cases are indicated in Fig. 14 by the matrix areas framed by solid lines. The same change process may be visible for a longer time interval and so present at multiple image pairs, as indicated in Fig. 14 by the matrix areas framed by dashed lines. For instance, the red dashed area covers the combination of the plant growing process with the previous seedbed preparation and the subsequent plant drying process. In these areas, the obtained change representation will depend on how the combination of these physical changes is perceived by the sensor, according

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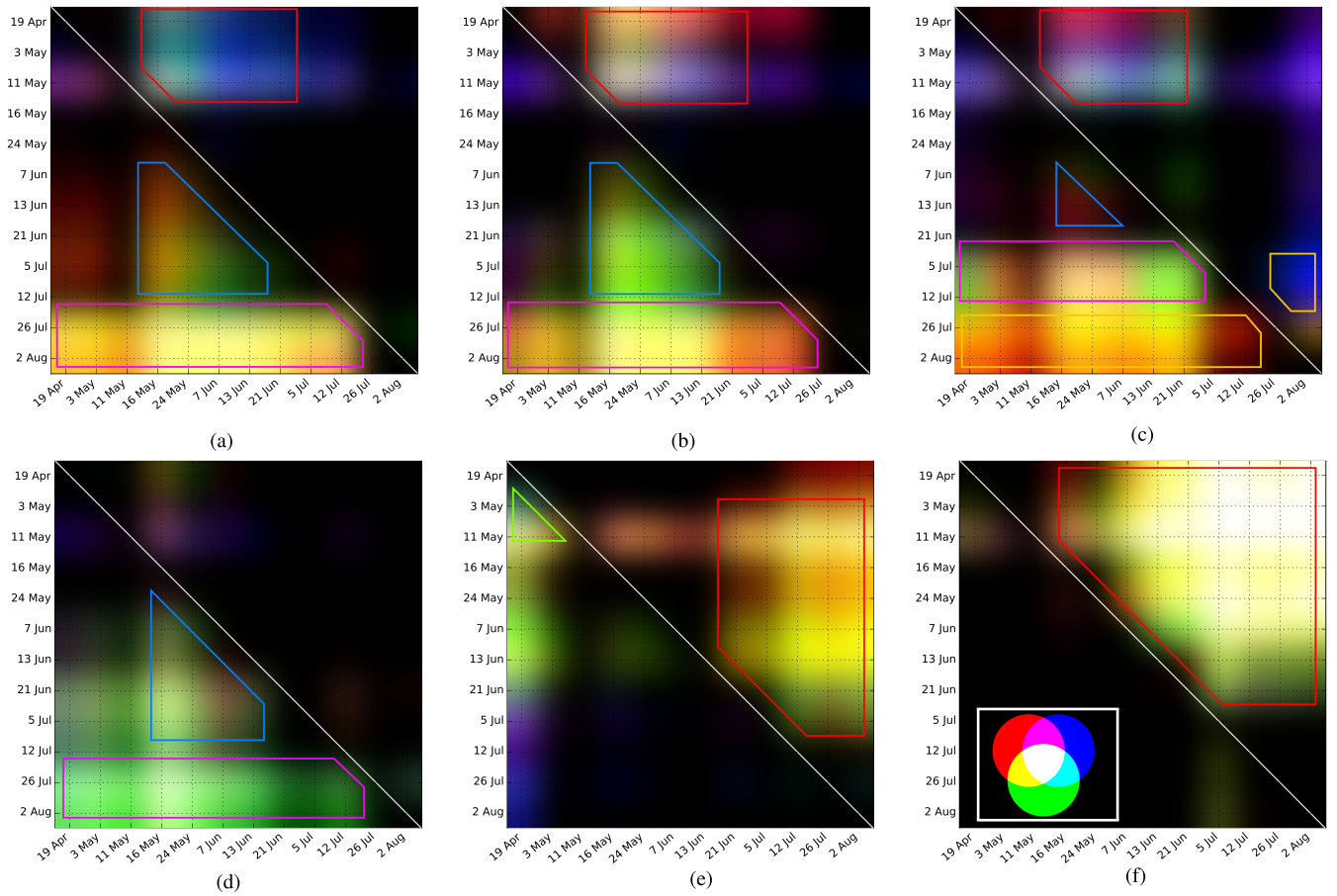

Fig. 15. Change matrices for different types of crops. The upper diagonal represents the increase $\mathbf{p}_{\text{inc}}$, and the lower diagonal represents the decrease $\mathbf{p}_{\text{dec}}$ between each acquisition pair. The visible changes of those described in Fig. 14 are marked, employing the same colors (a) Wheat, field 292. (b) Wheat, field 391. (c) Barley, field 440. (d) Rape, field 101. (e) Maize, field 222. (f) Sugar beet, field 460.

to the multiplicative combination of the different $\mathbf{M}$ matrices, as described in (16). Then, the obtained results in these areas will depend on the crop type, the plant geometry and density, the radar penetration, and so on.

In the case of the AgriSAR 2006 campaign, all changes described in Fig. 14 will not be visible for each crop type as the campaign does not cover the whole phenological cycle of all crops, as shown in Fig. 4. However, the temporal shift between the phenological cycles of the different crops, caused by the fact that some crops are planted before others, allows the detection of each change for at least one crop type.

Fig. 15 shows the change matrices for six fields: two wheat fields, barley, rape, maize, and a sugar beet field. The observed changes from those described in Fig. 14 have been labeled employing the same colors for easy identification in all the change matrices. For wheat, the selected two fields correspond to the two different changing behaviors appearing in Fig. 10(a). In Fig. 15, the scaling of $\mathbf{p}_{\text{inc}}$ and $\mathbf{p}_{\text{dec}}$ on the change matrices is between 1 and 8 dB.

As it may be seen, the two different wheat growing changes observed in Fig. 10(a) may also be seen in Fig. 15(a) and (b), especially in the first row, which corresponds to the comparison of the first acquisition, on the 19th of April, with the rest of acquisitions, as the change analyzed in Section IV-A. Apart from these small color changes, the change matrices for the two wheat fields are very similar, having the same change regions associated with the growing period, the fruit maturation, and the plant drying, as labeled in Fig. 15(a) and (b). Comparing Fig. 15(a)–(c), it may be observed that the wheat and barley crops present similar growing periods, but the drying process (marked in blue) is faster for the barley, as shown in Fig. 15(c). For the barley, even the harvesting is observed at the end of the acquisition campaign, marked in yellow color. Interestingly, the harvesting phase appears as a combination of a decrease in the volume scattering due to the elimination of the plant structure and an increase in the ground component. Therefore, the decrease and increase observed due to the harvest appear with different colors, indicating a different type of change.

In the rape field, as discussed in Section IV-A, the plant growth is not observed due to the incapacity of the radar to penetrate to the ground. However, the later changes in the vegetation due to the fruit maturation and plant drying clearly appear in the change matrix, as marked in Fig. 15(d). For the maize and sugar beet crops, as shown in Fig. 15(e) and (f), which are planted later, only the growing period is observed, as marked by a red polygon. In addition, in the maize field, even the seedbed preparation and planting may be seen in the first acquisitions, labeled in green.

It is worth noticing that for most of the cereal crops, including wheat, barley, and maize, the change corresponding to the growing period, labeled in red, starts around the stem

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elongation phase, equivalent to 30 s in the BBCH scale, as depicted in Fig. 4. However, for the sugar beet crop, this change is detected much earlier, in the leaf development phase, which corresponds to 10 s in the BBCH scale. This could be produced by the fact that the sugar beet leaves are bigger, having a higher surface than those of the cereals, or also due to the higher water content of these crops. In any case, this exemplifies how the changes and evolution of the different crop types are perceived unevenly by the radar.

Looking at the different change matrices presented in Fig. 15, it may be seen that they are distinct for every type of crop, as every crop follows a different temporal evolution. Even the two distinct changes for the wheat fields, as shown in Fig. 15(a) and (b), are very similar between them. Then, the change matrix may be seen as a crop signature, which may be very useful for crop identification and land use classification.

Apart from the big generic change processes labeled according to the scheme in Fig. 14, some small changes or dots may also be observed. This is especially the case for the acquisition of the 11th of May since smaller changes may be observed on this date for some of the change matrices. This was a particularly dry acquisition as no rain was measured during the two weeks before. Hence, a faint spot is observed as a decrease in the change matrix, especially for the species with lower vegetation height, as wheat, barley, and sugar beet. However, since this change is an isolated fact occurring only in one acquisition, the spot observed in the change matrices is not continued in time, as opposed to the other changes observed due to the crop evolution and marked in the different colors. Therefore, sufficiently dense time series are desired in order to be able to separate these isolated cases from the rest of the temporal evolution.

V. Conclusion

A new change detection technique able to quantify and interpret changes in multitemporal polarimetric SAR acquisitions has been proposed. It is based on the generalized eigendecomposition of the covariance matrices of the individual acquisitions. The generalized eigenvalues are related to the polarimetric contrast ratio and, thus, to the maximum and minimum changes between the covariance matrices. The generalized eigenvectors represent the polarization states (i.e., scattering mechanisms) associated with these changes. This allows not only the estimation of the change magnitude but also the physical interpretation of the occurred changes. Weighting the scattering mechanisms by the amount of their change allows the visualization and interpretation of the change information in terms of two Pauli change images that comprise the type of change in the color information and the magnitude of change in the intensity.

The proposed technique has been employed and tested for monitoring the phenological development of agricultural vegetation on the multitemporal AGRISAR 2006 L-band campaign data. The analysis is focused on five important crop types: wheat, barley, rape, maize, and sugar beet. For the different crop types, the proposed technique was able to detect and interpret a number of changes related to crop development as the growing, maturation, and drying processes. Even for the same crop type, for example, the winter wheat crops, a different change behavior can be observed in different fields. This is related to a temporal shift of the phenological cycle in the different fields, as, in some fields, the heading process started earlier than on others.

For better visualization and interpretation of the changes occurring in long time series, the change matrix representation has been introduced. It has proven to be able to indicate and interpret changes related to important development stages and/or transitions between them, including the seedbed preparation, planting, growing, fruit development, senescence, and harvest. The results indicate the potential of this approach for monitoring the temporal evolution of agricultural plants. It is worth mentioning that the interpretation of the change matrices is strongly dependent on the frequency at which the SAR acquisitions are made and it may be different for other frequencies.

Finally, it is worth to mention that the proposed polarimetric change analysis uses only a subset of the parameters provided by the generalized eigendecomposition, namely, the intensities of the changing polarization states. The associated phase information remains unused and may be further exploited in the future.

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